Notes for C++

Section 0 : Introduction

Who should do the course?

Students who want to major in any one (or two) of Computer Science, Data Science, Information Technology, Information Systems, Software Engineering, Games Programming (minor).

What do I need to know before I start the course?

No experience is required. Students with considerable programming experience in C++, Java, Python or a similar language can apply to skip the course. For more information, contact the Academic Dean Information Sciences. Note that skipping a course does not give you credit for the course. It simply means that you room in your degree to fit in a different course.

What is an Algorithm?

An algorithm is a series of instructions (and may have nothing to do with a computer).
Algorithms have been used for centuries to provide precise, detailed instructions.
Algorithms are written (or spoken) in *natural languages*, e.g. English is a natural language.

Here is an example of an algorithm (nothing to do with a computer):

1. Preheat the oven to 375 F
2. Mix the flour, baking powder, salt and sugar in a large bowl
3. Add the egg, milk and butter, stirring only enough to dampen the flour – the batter should not be smooth.
4. Stir 1 cup of blueberries into the batter.
5. Spoon into muffin pans, filling each cup about two-thirds full.
6. Bake for about 20 to 25 minutes.

Are these instructions precise enough? What could go wrong? For some ideas on the difficulties of getting an algorithm correct, have a look at https://www.youtube.com/watch?v=cDA3_5982h8

What is a Computer Program?

A computer program is a series of instructions written in a particular *programming language*. A program consists of instructions (statements) written in a specific form that a computer is able to translate and execute. Execute means to carry out the instructions.

Programming Languages

There are about 700 programming languages. Different languages were created to do different types of programming tasks. For example, the Structured Query Language (SQL) was developed to manage data held in a database. A language provides rules for writing instructions. In this course we use the language C++ which forms the basis of a family of related languages including C, C#, C++ and Java. (Interesting fact – there are 700 programming languages but there are over 6,500 spoken languages.)

Syntax

All languages including ***programming languages*** (such as C++) and ***natural languages*** (such as English) have rules on how to properly construct sentences (or equivalent) in the language. These rules are known as the **syntax** of the language. Two examples of syntax rules for English are (a) start every sentence with a capital letter, and (b) a question must end with a question mark. The full set of syntax rules for a language is called a ***grammar***. Note that English has sentences but programming languages have ***statements***.

People are amazing. They have an ability to read very badly constructed sentences and (usually) make sense of them. For example, a friend sent me the message "How R U?". I pointed out to my friend that the message is rubbish, it is not ***syntactically correct***. In other words, it does not follow the rules of English. My friend was surprised that I could not understand the message.

People are way better than computers. If you want to send an instruction to a computer it has to be 100% syntactically correct. Even 99% correct is not good enough. The computer will only understand the instruction if it is 100% correct. It is therefore very important, as a programmer, to be completely familiar with constructing correct statements in a program. The only way to get good at this is by practice. Always keep practicing your programming.

Our programming course includes learning the various types of statements that are available in the C++ language. Each time we introduce a new statement we discuss the syntax (the rules) of how to properly construct that type of statement.

Here is the first syntax rule for C++ and you will use this rule in every program:
A statement in C++ must end with a semi-colon

Semantics

Semantics refers to the ***meaning*** of a sentence in the language (or the logic). Note that perfect syntax does not imply good semantics. For example, the following English sentence has perfect syntax (it follows the rules of English) but has no meaning at all:

"Colourless green ideas sleep furiously" (*Noam Chomsky, 1957*)

Each time we learn a new ***statement*** in C++, we will learn the syntax for the statement (how to write it) and the semantics for that statement (what happens when the computer executes the statement).

The Editor and Compiler for this course

Editors are used to enter program instructions. Compilers are used to translate instructions to machine code. There are a large number of free editors and compilers available for C++ and if you are already using a particular editor/compiler, feel free to continue using it.

We will provide a free, downloadable editor and compiler for you to use.

Extra reading (don't worry – it's not going to be in the exam!)

Find out more about algorithms:
http://computer.howstuffworks.com/question717.htm

Find out more about computer programming:

http://en.wikipedia.org/wiki/Computer_programming

Find out more about programming languages:
https://en.wikipedia.org/wiki/Programming_language

Some information about C++:
https://www.learncpp.com/cpp-tutorial/introduction-to-cplusplus/

Extra online practice (only if you want to, this is not part of our course)

This is a good website to help you to learn C++ programming:
https://www.learn-cpp.org/

We cover most of what is in "the Basics". Warning – this is not our course, it may cover things slightly differently from our course, it is only to get some extra practice.